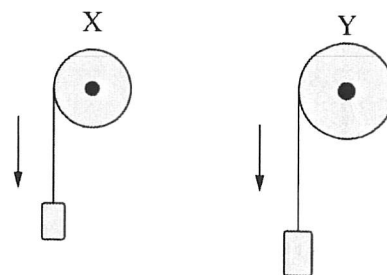
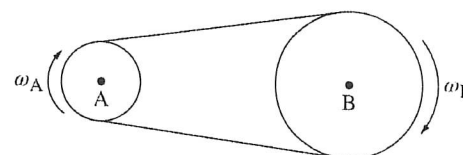


- A 1. Pulleys X and Y are each attached to a block by a string that wraps around the pulley. Both blocks are released from rest and have the same linear acceleration  $a$ . As the blocks fall, the pulleys rotate about their centers. Pulley Y has a larger radius than Pulley X. How does the final angular velocity  $\omega_Y$  of Pulley Y compare to the final angular velocity  $\omega_X$  of Pulley X?



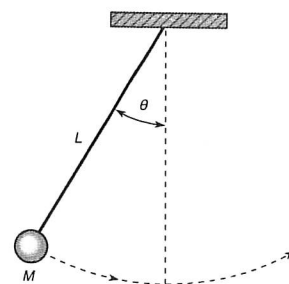
- (A)  $\omega_Y > \omega_X$
- (B)  $\omega_Y < \omega_X$
- (C)  $\omega_Y = \omega_X$
- (D) It cannot be determined without knowing the relative masses of each block.

- C 2. Wheels A and B are connected by a moving belt and are both free to rotate about their centers, as shown. The belt does not slip on the wheels. The radius of Wheel B is three times larger than the radius of Wheel A. Wheel A has constant angular speed  $\omega_A$  and Wheel B has constant angular speed  $\omega_B$ . Which of the following correctly relates  $\omega_A$



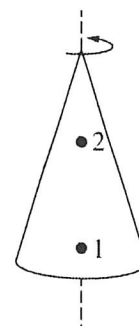
- (A)  $\omega_A = \frac{\omega_B}{3}$
- (B)  $\omega_A = \omega_B$
- (C)  $\omega_A = 3\omega_B$
- (D)  $\omega_A = 6\omega_B$

- P 3. A pendulum is formed from a mass  $M$  connected to a string of length  $L$ . The pendulum is initially raised to form an angle  $\theta$  (measured in radians) and released. The time it takes the pendulum to swing to the opposite side and return to its initial position is measured to be  $T$  (measured in seconds). Which of the following is a correct expression for the average linear speed of the pendulum?



- (A)  $\frac{\theta}{T}$
- (B)  $\frac{\theta L}{T}$
- (C)  $\frac{2\theta L}{T}$
- (D)  $\frac{4\theta L}{T}$

- C 4. A solid, uniform cone is spinning with constant angular speed about a vertical axis through its center of mass, as shown. A small insect is initially standing on the surface of the cone at Point 1. At a later time, the insect moved to Point 2. Which of the following statements about the motion of the insect is true?



- (A) The angular speed of the insect is greater at Point 1 than at Point 2.
- (B) The angular speed of the insect is greater at Point 2 than at Point 1.
- (C) The linear speed of the insect is greater at Point 1 than at Point 2.
- (D) The linear speed of the insect is greater at Point 2 than at Point 1.

## B Side

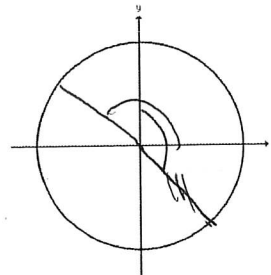
1. A disk has a radius of  $r = 2.5 \text{ m}$  and undergoes an angular displacement of  $3\pi/4$  radians.

- a. Sketch this angular displacement  $\frac{3}{4}\pi$   
 b. Convert this measurement to degrees.

$$135^\circ$$

- c. Determine the arc length formed.

$$\frac{135}{180} \times 5 \times \pi = 11.78 \text{ m}$$



2. A woman walks in a circular path of radius  $R = 20 \text{ m}$ . The woman undergoes an angular displacement of 2.2 rotations in the counterclockwise direction, which takes her 20 seconds to complete.

- a. Sketch this angular displacement  
 b. Convert her angular displacement to radians.

$$4.4\pi$$

- c. Calculate the woman's linear displacement.

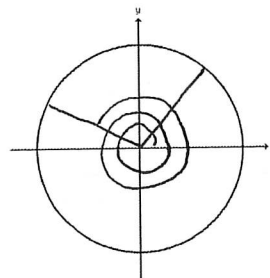
$$\pi \cdot D \cdot 2.2 = \pi \cdot 40 \times 2.2 \\ = 276.46 \text{ m}$$

- d. Calculate the woman's angular velocity (in rad/s)

$$\frac{4.4\pi}{20 \text{ s}} = 0.22 \text{ rad/s}$$

- e. Calculate the woman's linear velocity (in m/s)

$$\frac{276.46}{20} = 13.82 \text{ m/s}$$



3. A record has a small red dot on its edge. You measure that the dot travels a distance of 1.3 m (clockwise) when the record makes one full rotation, which takes 1.2 seconds.

- a. Determine the radius of the record

$$\frac{1.3}{\frac{\pi}{2}} = 0.2 \text{ m}$$

- b. Determine the angular velocity of the record (in deg/s)

$$\frac{360}{1.2 \text{ s}} = 300 \text{ deg/s}$$

